

What does the VLM See?

ML Team, Phronetic AI

June 2026

Abstract

Video Large Language Models process all visual tokens uniformly across every transformer layer, regardless of their contribution to the final answer. The resulting inference cost scales quadratically with sequence length, making deployment under tight compute budgets challenging. Existing token reduction methods operate at one of two levels: frame selection before the model, or attention-based pruning inside the model. Frame-level methods such as VideoITG (Shihao Wang et al. 2026) and PEEK (Steunou et al. 2026) require either a text query at selection time or reference caption supervision, limiting their generality. Token-level methods such as (Huang and Y. Li 2026, DyCoke (Tao et al. 2025, either operate without any language grounding (remaining agnostic to what the model actually needs), or derive supervision from task-specific reward signals that are expensive to compute. We take a different position: the frozen VLM itself contains a reliable, model-native supervision signal for token importance, localisable to a small subset of intermediate layers. Recent mechanistic analysis of Video-LLMs (Gou et al. 2025) demonstrates that language-to-video attention at layers 12–16 accounts for the disproportionate majority of visual information retrieval, while temporal and spatial self-attention between video tokens contributes comparatively little. Crucially, this signal is distinct from the CLS or BOS attention exploited by earlier pruning methods (such as FastV), which are known to exhibit positional bias and inflate importance scores for semantically empty tokens (Myrzakhan et al. 2026). Language-to-video attention at critical intermediate layers does not exhibit this bias Takezoe et al. 2026. We train a lightweight MLP scorer (300K parameters) online, supervised by language-to-video attention extracted from layers 12–16 of a frozen Qwen2.5-VL during forward passes on standard

VideoQA data. To prevent temporal coverage collapse - where globally sorting top tokens concentrate in a single visual moment, we apply stratified selection across non-overlapping temporal bins ensuring the retained set spans the full video. Spatial and temporal identity of the token is preserved via RoPE, so the model always knows where and when each token originated from. Only the scorer is trained; VLM remains frozen.

Our contributions are: (1) a training signal derived from model-internal language-to-video attention at empirically validated critical layers, requiring no caption alignment, no reward model, and no architectural modification; (2) pre-projector token gating, saving compute at an earlier stage than post-projection pruning methods; (3) stratified temporal selection ensuring coverage-aware retention without merging or architectural overhead; and (4) an empirical finding that norm-based heuristics reliably identify uninformative tokens but fail to identify important ones, directly motivating the need for a learned selector. This work extends our prior result (Dave et al. 2026) that motion-based heuristics suffice for video compression but are structurally insufficient for understanding

1 Introduction

The dominant paradigm in video understanding has shifted decisively toward large multimodal models. Video Large Language Models (Video-LLMs) now achieve strong performance across a wide range of tasks such as video question answering, temporal reasoning, dense captioning done by processing sampled frames as sequences of visual tokens alongside a language query. Yet this capability comes at substantial computational cost. The quadratic scaling of attention with sequence length means that visual tokens alone account for the overwhelming majority of inference FLOPs, often exceeding language token compute by two orders of magnitude (Gou et al. 2025).

The standard response to this bottleneck has taken two forms. The first is frame-level selection: choose which frames to show the model before any encoding occurs. Methods in this family, including PEEK (Steunou et al. 2026) and VideoITG (Shihao Wang et al. 2026), have demonstrated that adaptive frame selection outperforms uniform sampling, particularly at low frame budgets. However, they operate on decoded pixels via external vision-language models, their supervision is tied to reference captions or text queries

available only at selection time, and they discard entire frames rather than making fine-grained spatial decisions within frames.

The second response is token-level pruning inside the model: retain all frames but reduce the number of tokens that propagate through the transformer layers. Training-free methods such as KiToke (Huang and Y. Li 2026) and DyCoke (Tao et al. 2025) reduce spatiotemporal redundancy via diversity maximisation or cosine similarity thresholding. These approaches are principled and require no training, but they are query-agnostic, optimising for visual diversity without reference to what the model actually attends to when answering a question. Training-based methods such as (Shida Wang et al. 2026) learn a selection policy via reinforcement learning, using task accuracy as a reward signal. This produces a task-grounded selector but at substantial training cost: each update requires multiple full VLM forward passes to evaluate sampled token masks.

We argue that both families miss a simpler and more direct source of supervision: the VLM itself. Recent mechanistic analysis Gou et al. 2025 shows that language-to-video attention at a small subset of intermediate layers, specifically layers 12–16 in 3B to 7B-scale models which has shown to account for the disproportionate majority of visual information retrieval in VideoQA. Most other layers contribute negligibly; temporal and spatial self-attention between video tokens is largely redundant. This finding implies that the model already "knows" which tokens matter, and that knowledge is readable from its own attention patterns at a well-defined depth.

We exploit this directly. We train a lightweight MLP scorer ($\sim 300\text{K}$ parameters) online, supervised by language-to-video attention extracted from layers 12–16 of a frozen Qwen2.5-VL. The scorer operates on visual token embeddings before the projector — gating tokens at the earliest possible stage, before any LLM computation occurs. Crucially, we address a failure mode common to global top-k selection: without temporal constraints, high-scoring tokens can concentrate in a single visually rich moment, leaving later frames entirely unrepresented. We resolve this with stratified temporal selection, distributing the token budget across non-overlapping temporal bins while the scorer determines which tokens within each bin to retain. Spatial and temporal positional identity is preserved through M-RoPE position IDs, so the VLM retains full knowledge of where and when each surviving token originated.

This work connects directly to our prior contribution (Dave et al. 2026), which demonstrated that simple L1 delta thresholding over continuous video

latents suffices for high-quality reconstruction without learned routing. That result showed motion-based heuristics are sufficient for compression. Here we show they are insufficient for understanding: norm-based heuristics reliably discard uninformative tokens (bottom-k selection is stable) but fail to identify semantically important ones (top-k selection degrades prediction consistency). This asymmetry directly motivates the learned scorer proposed here.

2 Method

Our method consists of three components: a lightweight token scorer trained online on a frozen VLM’s internal attention, a stratified temporal selection strategy that enforces coverage across the video, and a pre-projector gating mechanism that removes low-scoring tokens before any LLM computation occurs.

2.1 Preliminaries

A Video-LLM typically consists of a frozen visual encoder Enc_v , a projector Proj , and a frozen decoder-only language model with L transformer layers. Given a video V sampled at T frames, the visual encoder produces patch embeddings:

$$F = \text{Enc}_v(V) \in \mathbb{R}^{T \times N \times d_v} \quad (1)$$

where N is the number of patches per frame and d_v is the visual encoder hidden dimension. These are projected into the LLM token space via Proj , concatenated with text tokens, and processed through L transformer layers. The full sequence length $T \cdot N + T_{\text{text}}$ drives the quadratic attention cost, motivating reduction of $T \cdot N$ before any LLM computation.

2.2 Supervision Signal: Language-to-Video Attention

We derive per-token importance scores from the frozen VLM’s internal attention at a set of critical layers $\mathcal{L} = \{12, 13, 14, 15, 16\}$, identified empirically by Gou et al. (2025) as the layers where language-guided visual retrieval is concentrated.

At each critical layer $\ell \in \mathcal{L}$, the attention module produces a weight matrix $A^{(\ell)} \in \mathbb{R}^{H \times S \times S}$, where H is the number of attention heads and $S =$

$T \cdot N + T_{\text{text}}$ is the full sequence length. We extract the language-to-video slice i.e., query positions corresponding to text tokens, key positions corresponding to video tokens and aggregate:

$$s_i = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \frac{1}{H \cdot T_{\text{text}}} \sum_{h=1}^H \sum_{j=1}^{T_{\text{text}}} A_{h, T \cdot N + j, i}^{(\ell)} \quad (2)$$

This yields a scalar importance score s_i for each video token $i \in \{1, \dots, T \cdot N\}$. Scores are min-max normalised per sample to $[0, 1]$. During training, the VLM forward pass is truncated at layer 16 and attention weights are detached; no gradients flow into the VLM.

We use text-to-vision attention specifically because it is resistant to the positional bias and attention sink effects that afflict vision encoder CLS attention and early-layer LLM attention. Vision-to-vision attention exhibits consistent recency bias; text-to-vision attention in middle layers does not (Takezoe et al. 2026).

2.3 Token Scorer

The scorer is a two-layer MLP with LayerNorm input that maps each visual token embedding to a scalar importance logit:

$$\hat{s}_i = W_2 \text{GELU}(W_1 \text{LN}(f_i) + b_1) + b_2 \quad (3)$$

where $f_i \in \mathbb{R}^{d_v}$ is the ViT patch embedding for token i , $W_1 \in \mathbb{R}^{d_h \times d_v}$, $W_2 \in \mathbb{R}^{1 \times d_h}$, and $d_h = 256$. The scorer has approximately 300K trainable parameters. The frozen visual encoder, projector, and all LLM layers are never updated.

Training Objective. Token selection is fundamentally a ranking problem: only the relative order of importance scores determines which tokens survive, not their absolute values. We therefore adopt the ListMLE list-wise ranking loss (Lan et al. 2014). Let π^* denote the permutation of token indices sorted by decreasing teacher score s_i . The loss is:

$$\mathcal{L}_{\text{ListMLE}} = - \sum_{i=1}^{T \cdot N} \log \frac{\exp(\hat{s}_{\pi^*(i)})}{\sum_{\tau=i} \exp(\hat{s}_{\pi^*(\tau)})} \quad (4)$$

This is the Plackett-Luce negative log-likelihood of observing the teacher-induced ranking under the predicted logits, optimising the full ranking jointly rather than independently per token.

Online Training. At each training step, a video-question-answer triple is sampled from a VideoQA dataset. A single forward pass through the frozen VLM to layer 16 with `output.attentions=True` produces the supervision signal s . The scorer forward pass produces $\hat{s}$. The ListMLE loss is computed and gradients are propagated through the scorer only. No labels beyond the standard VideoQA annotations present in the dataset are required.

2.4 Stratified Temporal Selection with Pareto-Adaptive Budgeting

A global top- k selection strategy applied to scorer logits risks temporal coverage collapse: tokens from visually rich frames accumulate disproportionately, leaving later or quieter frames entirely unrepresented. Conversely, a uniform per-bin allocation $k_t = \lfloor K/T \rfloor$ wastes token budgets on static, uninformative background frames.

We resolve this by setting k_t adaptively, exploiting the empirical observation that the tail of visual token importance follows a Pareto distribution, whose shape parameter α dictates the concentration of information. To properly estimate this, we must first establish the optimal estimator for a Pareto tail.

The Naïve Model and its Maximum Likelihood Estimator. Assume a set of token norms $x_i \stackrel{\text{iid}}{\sim} \text{Pareto}(\alpha, x_m)$ with density $f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$ for $x \geq x_m$. The joint likelihood function for n observations is the product of their individual densities:

$$L(\alpha) = \prod_{i=1}^n \frac{\alpha x_m^\alpha}{x_i^{\alpha+1}} \quad (5)$$

Taking the natural logarithm transforms this product into a sum, giving the log-likelihood:

$$\ell(\alpha) = \sum_{i=1}^n \log \left(\frac{\alpha x_m^\alpha}{x_i^{\alpha+1}} \right) = n \log(\alpha) + n\alpha \log(x_m) - (\alpha + 1) \sum_{i=1}^n \log(x_i) \quad (6)$$

To find the maximum likelihood estimator (MLE), we take the derivative with respect to α and set it to zero:

$$\frac{\partial \ell}{\partial \alpha} = \frac{n}{\alpha} + n \log(x_m) - \sum_{i=1}^n \log(x_i) = 0 \quad (7)$$

Solving for α and pulling the $n \log(x_m)$ term into the summation yields the exact MLE:

$$\hat{\alpha}_{\text{MLE}} = \frac{n}{\sum_{i=1}^n \log(x_i/x_m)} \quad (8)$$

This formulation is statistically robust because the transformation $Y = \log(X/x_m)$ yields an Exponential distribution. By evaluating the Pareto CDF at $x_m e^y$, we find $F_Y(y) = 1 - (x_m/x_m e^y)^\alpha = 1 - e^{-\alpha y}$, which is exactly $Y \sim \text{Exp}(\alpha)$. Consequently, the MLE for α inherits the standard properties of an exponential rate estimator, including a known exact variance and inverse-Gamma sampling distribution.

Extreme Value Theory and the Hill Estimator. However, directly applying $\hat{\alpha}_{\text{MLE}}$ to the full distribution of ViT activation norms commits a model misspecification. The bulk of the distribution is approximately log-normal; only the extreme upper region—where the semantically important tokens live—is Pareto. Fitting the naïve model to the whole sample allows the bulk to dominate, resulting in inflated and unstable α estimates.

Following Extreme Value Theory, we relax the assumption to a regularly varying tail: $\Pr(X > x) = x^{-\alpha} L(x)$, where $L(x)$ is a slowly varying function. Given a sufficiently high threshold u , the scaled exceedances X/u are approximately $\text{Pareto}(\alpha, 1)$. We can therefore adapt our derived MLE to operate strictly conditionally on the top k order statistics. Substituting n with k , the threshold x_m with $x_{(n-k)}$, and evaluating over the top k values $x_{(n-i+1)}$, we obtain the conditional MLE, whose inverse is the well-known Hill estimator:

$$\hat{H}_{k,n} = \frac{1}{\hat{\alpha}} = \frac{1}{k} \sum_{i=1}^k \log \frac{x_{(n-i+1)}}{x_{(n-k)}} \quad (9)$$

Rényi Representation and Finite-Sample Properties. To rigorously analyze the bias-variance trade-off in selecting the threshold k , we must establish the exact finite-sample distribution of the Hill estimator. As established previously, for an exact Pareto tail, the transformation $Y = \log(X/x_{(n-k)})$

yields $Y \sim \text{Exp}(\alpha)$. Because α acts as a rate parameter, we can express this as a scaled standard exponential: $Y = E/\alpha$, where $E \sim \text{Exp}(1)$. Substituting this scaling into the Hill estimator translates the components into differences of standard exponential order statistics:

$$Y_{(n-i+1)} - Y_{(n-k)} = \frac{1}{\alpha} (E_{(n-i+1)} - E_{(n-k)}) \quad (10)$$

Here, we invoke the Rényi representation theorem. Due to the unique memorylessness property of the exponential distribution, the excesses of the top k order statistics over the threshold $E_{(n-k)}$ are strictly independent of the threshold itself. Furthermore, these k gaps are jointly distributed exactly as the order statistics of a fresh, independent sample of k standard exponentials, denoted $E'_j \sim \text{Exp}(1)$. Since summing a set of order statistics is mathematically identical to summing the unsorted variables, the Hill estimator simplifies remarkably in distribution:

$$\hat{H}_{k,n} \stackrel{d}{=} \frac{1}{\alpha k} \sum_{j=1}^k E'_j \quad (11)$$

This elegant reduction frames the inverse Hill estimator simply as $1/\alpha$ scaled by the sample mean of k iid standard exponential variables. Given $\mathbb{E}[E'_j] = 1$ and $\text{Var}(E'_j) = 1$, we can isolate the exact finite-sample expectation and variance under a strict Pareto tail:

$$\mathbb{E}[\hat{H}_{k,n}] = \frac{1}{\alpha k} \sum_{j=1}^k \mathbb{E}[E'_j] = \frac{1}{\alpha} \quad (12)$$

$$\text{Var}(\hat{H}_{k,n}) = \frac{1}{\alpha^2 k^2} \sum_{j=1}^k \text{Var}(E'_j) = \frac{1}{\alpha^2 k} \quad (13)$$

This exact variance cleanly highlights the central tension in threshold selection. To drive the variance of the estimator to zero, we require $k \rightarrow \infty$. However, as k increases, the threshold $x_{(n-k)}$ descends into the non-Pareto bulk of the distribution, violating the slowly varying assumption of $L(x)$ and introducing massive systematic bias. Thus, temporal budgeting requires finding the optimal plateau where k is sufficiently large to stabilize the $1/(\alpha^2 k)$ variance, yet strictly constrained such that $k/n \rightarrow 0$ to avoid bulk contamination.

2.5 Pre-Projector Gating and Positional Identity

The retained embeddings $F_{\mathcal{I}} = \{f_i : i \in \mathcal{I}\} \in \mathbb{R}^{K \times d_v}$ are passed to the frozen projector. All other tokens are discarded before any LLM computation occurs, saving projector and full-depth LLM compute proportional to $1 - K/(T \cdot N)$.

Critically, each retained token carries its original spatiotemporal position identifiers namely the frame index t , spatial row x , and spatial column y , into the projector and through the LLM via M-RoPE (Bai et al. 2025). The VLM therefore retains full knowledge of where and when each surviving token originated, without any positional re-indexing. The full inference pipeline is:

$$V \xrightarrow{\text{Enc}_v} F \xrightarrow{\text{Scorer}} \hat{s} \xrightarrow{\text{Stratified top-}k} F_{\mathcal{I}} \xrightarrow{\text{Proj}} \text{Tokens} \xrightarrow{\text{VLM}} \text{Answer} \quad (14)$$

3 Proposed Experiments

3.1 Setup

We evaluate on three VideoQA benchmarks spanning different temporal scales and task types:

- **VideoMME** (Fu et al. 2025) - 900 videos across six domains with short, medium, and long duration splits. Primary benchmark for overall VideoQA accuracy.
- **MVBench** (K. Li et al. 2024) - 4,000 QA pairs across 20 temporally-sensitive task types including action sequences, object permanence, and motion analysis.

The base VLM throughout is Qwen2.5-VL-7B-Instruct. The scorer is trained on the NExTVideo training split (**nextqa**) for 10,000 steps with AdamW (lr = 10^{-4} , weight decay 10^{-2}), cosine annealing schedule, and gradient clipping at 1.0.

3.2 Baselines

We compare at token retention ratios $\rho \in \{25\%, 50\%, 75\%\}$ of the full token count $T \cdot N$:

- **Full model** - no token reduction, upper bound.
- **Uniform sampling** - spatially and temporally uniform subsampling, strong baseline (Steunou et al. 2026).
- **L1 delta thresholding** - our prior BMVC heuristic adapted to token-level selection, representing the motion-based approach.
- **KiToke** (Huang and Y. Li 2026) - best published training-free token compression at matched retention ratios.
- **SCORE** (Shida Wang et al. 2026) - best published training-based token compression, representing RL-supervised selection.
- **Ours** - learned scorer with stratified temporal selection.

3.3 Primary Metrics

VideoQA accuracy (multiple-choice top-1) is the primary metric across all benchmarks. We additionally report wallclock inference speedup relative to the full model at each retention ratio, measured on a single Ampere architecture GPU with batch size 1.

3.4 Ablations

Supervision source. We ablate the supervision signal by replacing language-to-video attention with: (a) uniform random targets, (b) CLS attention from the vision encoder, and (c) all-token average attention. This directly tests whether the choice of critical-layer language-to-video attention is the source of any gains over baselines.

Layer range. We vary $\mathcal{L}$ across $\{1-5\}$, $\{6-11\}$, $\{12-16\}$, and $\{17-28\}$ to verify that the critical layer range identified by Gou et al. (2025) is indeed the most informative for our supervision signal.

Selection strategy. We compare global top- k against stratified top- k to quantify the benefit of temporal coverage enforcement, particularly on long-form benchmarks where temporal collapse is most harmful.

Norm heuristic vs. learned scorer. We include a norm-based selector (top- k by L2 norm of ViT features) as an additional ablation, connecting directly to our empirical finding that bottom- k norm selection is reliable but top- k norm selection is not. This motivates the learned scorer within the paper’s own experimental narrative.

3.5 Expected Outcomes

We expect the learned scorer with stratified selection to outperform all training-free baselines (Uniform, L1 delta, KiToke) at low retention ratios ($\rho = 25\%$), where the quality of token selection matters most. The gap over SCORE should emerge from training efficiency: our method requires a single truncated VLM forward pass per training step, whereas SCORE requires K full forward passes. At $\rho = 75\%$, differences between methods are expected to narrow, as sufficient tokens remain for uniform coverage to be competitive — a pattern consistent with findings in **steunou2026peek** and **learnpruner2026**.

The ablation on selection strategy is expected to show the largest gains for stratified over global top- k on EgoSchema, given its three-minute video length and the higher probability of temporal collapse at aggressive retention ratios.

References

- Bai, Shuai et al. (2025). *Qwen2.5-VL Technical Report*. arXiv: 2502.13923 [cs.CV]. URL: <https://arxiv.org/abs/2502.13923>.
- Dave, Kevin et al. (2026). *Adaptive Tokenisation Via Temporal Redundancy Masking And Latent Inpainting*. arXiv: 2606.06158 [cs.CV]. URL: <https://arxiv.org/abs/2606.06158>.
- Fu, Chaoyou et al. (2025). *Video-MME: The First-Ever Comprehensive Evaluation Benchmark of Multi-modal LLMs in Video Analysis*. arXiv: 2405.21075 [cs.CV]. URL: <https://arxiv.org/abs/2405.21075>.
- Gou, Chenhui et al. (2025). *An Empirical Study on How Video-LLMs Answer Video Questions*. arXiv: 2508.15360 [cs.CV]. URL: <https://arxiv.org/abs/2508.15360>.

- Huang, Haifeng and Yang Li (2026). *KiToke: Kernel-based Interval-aware Token Compression for Video Large Language Models*. arXiv: 2604.03414 [cs.CV]. URL: <https://arxiv.org/abs/2604.03414>.
- Lan, Yanyan et al. (2014). “Position-aware ListMLE: a sequential learning process for ranking”. In: *Proceedings of the Thirtieth Conference on Uncertainty in Artificial Intelligence*. UAI’14. Quebec City, Quebec, Canada: AUAI Press, pp. 449–458. ISBN: 9780974903910.
- Li, Kunchang et al. (2024). *MVBench: A Comprehensive Multi-modal Video Understanding Benchmark*. arXiv: 2311.17005 [cs.CV]. URL: <https://arxiv.org/abs/2311.17005>.
- Myrzakhan, Aidar et al. (2026). *Sink-Aware Pruning for Diffusion Language Models*. arXiv: 2602.17664 [cs.CL]. URL: <https://arxiv.org/abs/2602.17664>.
- Steunou, Killian et al. (2026). *PEEK: Picking Essential frames via Efficient Knowledge distillation*. arXiv: 2605.31029 [cs.CV]. URL: <https://arxiv.org/abs/2605.31029>.
- Takezoe, Rinyoichi et al. (2026). *LearnPruner: Rethinking Attention-based Token Pruning in Vision Language Models*. arXiv: 2604.23950 [cs.CV]. URL: <https://arxiv.org/abs/2604.23950>.
- Tao, Keda et al. (2025). *DyCoke: Dynamic Compression of Tokens for Fast Video Large Language Models*. arXiv: 2411.15024 [cs.CV]. URL: <https://arxiv.org/abs/2411.15024>.
- Wang, Shida et al. (2026). *Dynamic Token Compression for Efficient Video Understanding through Reinforcement Learning*. arXiv: 2603.26365 [cs.CV]. URL: <https://arxiv.org/abs/2603.26365>.
- Wang, Shihao et al. (2026). *VideoITG: Multimodal Video Understanding with Instructed Temporal Grounding*. arXiv: 2507.13353 [cs.CV]. URL: <https://arxiv.org/abs/2507.13353>.